Introduction

We have studied earlier about the physical changes and chemical changes. Can you identify, from the following list which are physical changes and which are chemical changes?

- ❖ breaking of a stick into two pieces
- ❖ burning of a paper
- ❖ tearing paper into small pieces
- ❖ dissolving sugar in water
- ❖ burning of petrol or LPG gas
- ❖ water boiling into water vapour
- ❖ coconut oil becoming solid during winter

Can you see the important difference between the chemical change and physical change? When you cut a paper into two, both are still paper pieces, but once you burn it, there is no longer the paper, only some ash and the smoke are left.

Chemical change results in the change of the substance; In **physical change** only the shape, size or volume changes; the state of the matter may also change, from liquid to gas or from liquid to solid, however the substance remains, chemically as it is.

Let us do the following experiment. Add a pinch of turmeric powder to water; water turns yellow. Take a small quantity of soap water in a beaker and add a pinch of turmeric powder to it. Now, What happens? Is there any change in colour of the solution? Is it also turning to yellow or to some other colour?

Try adding turmeric powder to various household liquids and observe the result. Try it on, say, tamarind extract. Try it on with cleaning liquids in the house. Does it change the colour?

Chemists identify turmeric powder as a **'natural indicator'.** The change in colour indicates that the material is either acid or base medium.

Find answer for the following questions with the help of your teacher. This will help you to understand how chemistry plays vital role in our life.

- ❖ How does milk change into curd?
- ❖ How can you remove stain on the copper vessels?
- ❖ Idli is a little bit hard while we cook by using newly grinded idli dough but it is soft with old dough. Why?
- ❖ How does rusting of iron happen?
- ❖ Why does white sugar change into black when heating?

We can understand the chemical changes that happen around us by knowing the answers for the above questions.

6th Science_IIIrd Term English Book_Unit_3.indd 33 01-11-2019 5.39.43 PM